

Citation Under Pressure: What Deployment Constraints Do to Legal Citation Integrity, and What Deterministic Verification Can Repair

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Abstract. Sanctions over fabricated legal citations have made language-model drafting a problem for courts, and most research on the problem has produced detection benchmarks that score errors in finished briefs. This paper measures how the errors are produced. Six models from six vendors, from a 30B open-weight model to two frontier systems, drafted argument sections for 48 U.S. Supreme Court matters under five prompting conditions: baseline, a citation quota, a restriction to pre-1970 authority, a sanctions warning, and all three combined. Every citation in the 1,440 drafts was checked deterministically, with no language-model judge in any primary measurement: existence against an 18-million-row reporter index, quotations against a verbatim checker validated on seeded faults, and pinpoint pages against official reporter pagination from public archives. Under combined pressure the smallest model’s citation existence fell from 0.924 to 0.773, and verbatim quotation accuracy fell significantly for three of six models; only the strongest model was unaffected. In a second experiment the checker fed its findings back into revision. The strongest model reached 0.974 verbatim accuracy, and its scored quotations fell by more than half in the process while its count of accurate quotations barely moved; the rise came from deletion. Every model that could act on feedback lost accuracy under false feedback. We release the instruments, including a page-level pinpoint verifier built from public data.

Keywords. legal citation, hallucination, large language models, verification, evaluation

1. Introduction

Since 2023, courts have repeatedly sanctioned lawyers who filed briefs containing citations that a language model invented, beginning with *Mata v. Avianca* [1] and continuing in a growing docket that a public tracker now counts in the hundreds [2]. Most published work frames the problem as detection: given a brief that already contains errors, how many can a model or agent find [3,4]? Detection accuracy is the relevant measure for review of a filed brief. It does not tell a deployer which prompting conditions produce the errors, which is the question this paper addresses.

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This paper asks two production-side questions. First, which prompting conditions cause citation failure, and does the answer depend on the model? Prompts in practice carry constraints of the kind studied by Zhao et al. [5] for scholarly citations: a minimum count of authorities, a restriction on which authorities may be cited, and a warning about consequences. That study found that such constraints, temporal restrictions in particular, collapse citation existence even in capable models. We are not aware of a comparable study for law. Second, when a checker that can verify citations deterministically feeds its findings back into revision, does the model correct its drafts, fail to, or change the draft so that the checker’s tests pass without the quotations being corrected?

Both questions need an instrument the field has said does not exist in public. The best published detection agent misses nearly half of incorrect pinpoint citations (52.8 percent recall, with open-weight detectors between 19 and 26 percent), and the benchmark’s authors conclude that “no models can reliably detect wrong pincites” because page-level information lives behind commercial databases [3]. We built the instrument from public sources: citation existence and name pairing against a local index derived from CourtListener bulk data, verbatim quotation checking validated on seeded faults, and page-level pinpoint verification against the official reporter pagination in the Caselaw Access Project’s volume archives [6]. Every primary number reported here is produced by this instrument; no language-model judge is used in any primary measurement.

Our scope is the closed-book use of general-purpose models. The sanctioned filings documented to date were largely produced by practitioners and self-represented litigants using a general-purpose chatbot without retrieval [2,7]. Retrieval-grounded commercial tools are a different setting, and Section 7 says what we expect to transfer.

The paper reports three results. First, pressure degrades citation integrity beginning with the smallest model, and the degradation is rarely accompanied by refusal; in 1,152 pressured drafts there were three refusals. Second, feedback from a deterministic checker leads capable models to delete flagged quotations far more often than to correct them, and false feedback lowers the accuracy of every model that acts on feedback at all. Third, in the strongest models the remaining errors are mostly genuine legal language attributed to the wrong case or the wrong page, and the released instruments detect both classes.

2. Related Work

Legal citation reliability. Liu et al. [3] build a detection benchmark from real appellate briefs with injected errors and a taxonomy grounded in sanctioned filings; we adopt their five error categories as outcome coding and measure the naturally occurring analogue their limitations section asks for. LegalCiteBench [4] shows closed-book citation retrieval near zero and that failing models almost never abstain, which our pressure conditions reproduce in generation. Dahl et al. [7] measured hallucinated authority in general-purpose chatbots, and Magesh et al. [8] in commercial legal research tools; together they establish the rate of hallucinated authority in deployed systems.

Constraints and fabrication. Zhao et al. [5] is the closest design: five prompting regimes over scholarly claims, verified by fuzzy bibliographic matching that left 36 to 61 percent of citations unresolved. We transplant their factorial to law with deterministic verification (0.5 percent unresolved in our pilot) and a formal statistical model. Their mechanism, compliance with the temporal window filled with fabrications, transfers to law for some models and not others, and Section 4 reports which.

Revision with execution feedback. The loop experiment takes its protocol from execution-feedback work in code generation [9], including the control that separates feedback content from extra attempts. Our version adds a second control in which the feedback is false.

Language-model judges. The negative result on relevance judging joins evidence that agreement statistics overstate judge validity [10,11]; we add a calibration protocol with a pre-declared bar.

AI and law. The instability of model answers to legal questions is documented at ICAIL: Blair-Stanek and Van Durme [12] found that frontier models flip the winning party across identical zero-temperature runs, which motivates measuring reliability across a set of deployment conditions instead of a single prompt. Evaluation methodology in this community has moved toward separating factual truth from reasoning soundness [13] and pairing automated metrics with the checks lawyers perform [14]; citation integrity is a check for which deterministic ground truth is available, and this paper automates it. Closest to our pinpoint instrument, Garneau et al. [15] found that retrieving the paragraph a proposition rests on is far harder than retrieving the case, using network and text signals to propose pincites; our verifier approaches the same gap from the other side, checking a generated pincite against the page text. Zhang et al. [16] show models miss overruled precedent on 236 Supreme Court pairs, a failure class adjacent to the wrong-authority errors we measure. ConsRAG [17] mitigates hallucination through constrained retrieval; our loop experiment tests the complementary strategy of post-hoc deterministic feedback and finds a specific way it fails. Finally, Harasta et al. [18] report that lawyers rate identical documents lower when told a model wrote them; this paper proposes machine verification of citation integrity as a basis for trust that does not depend on disclosed provenance. Work on legal argument generation [19,20], citation-worthiness detection [21], argumentation over Supreme Court briefs [22], and citation-graph analysis [23,24,25] supplies the surrounding context.

3. Tasks, Conditions, and Instruments

Terms. An American case citation such as *Brown v. Board of Education*, 347 U.S. 483, 495 (1954) names the parties, the volume (347), the reporter (U.S., the official United States Reports), and the first page (483); the optional second page (495) is the *pincite*, which for a quotation must be the page where the quoted language appears. Throughout, *citation existence* is the share of adjudicable citations that resolve to a real case, and *strict quotation accuracy* is the share of scored quotations that reproduce the opinion verbatim. The five prompting conditions are called baseline, quota, temporal, stakes, and combined.

Matters. The matters are forty-eight Supreme Court cases from 1990 to 2020, drawn from a 500-case pool built from the Supreme Court Database [26]. Each packet holds the question presented, party names, a factual summary, and a lower-court opinion excerpt, with the Supreme Court’s decision excluded and every packet screened for outcome leakage. The model argues an assigned side (petitioner for even-numbered matters, respondent for odd) in an argument section of under 900 words. Nothing in the design predicts outcomes, which keeps the memorization confound of Supreme Court prediction tasks [27] out of scope.

Conditions. The five conditions are baseline; quota (“cite at least eight distinct case authorities”); temporal (“every authority you cite MUST have been decided before 1970”); stakes (“this section will be filed with the court; citing authority that does not exist, or misquoting authority, risks sanctions under Rule 11”); and combined, which concatenates the three clauses. Each condition has one fixed template, run at temperature 0. The stakes clause names Rule 11 of the Federal Rules of Civil Procedure because it is the sanctions rule invoked in the sanctioned-filing cases [1]. Rule 11 does not govern Supreme Court filings, which fall under the Court’s own rules and inherent authority; the clause functions as a stakes signal, and we make no claim about its legal accuracy. The conditions are designed to isolate mechanisms and are not intended to simulate practice. A robustness check re-ran twelve matters under alternative phrasings of every condition.

Models. The models are Qwen3-30B-A3B, Mistral Small, Llama-4 Maverick, DeepSeek V4 Flash, GPT-5.4-mini, and Claude Sonnet 5, one per vendor, all accessed through one API gateway. The hypotheses concern how effects change with capability, so the set spans open-weight 30B to frontier with no vendor contributing two models; we do not assume a total ordering, and tables are ordered by baseline strict quotation accuracy. A seventh model, GLM-4.7-flash, completed 207 of 240 drafts and is reported in Section 4.

Adjudication. Citations are extracted with *eyecite* [28] and given one of three labels: exists, not found, or unverifiable (vendor citations and coverage gaps), so that index coverage is never counted as fabrication; the unverifiable share is reported per cell in the released data. Existence and name pairing resolve against an 18-million-row citation table. Quotations are located by balanced quotation marks and attributed by rule: to the case named in the same sentence when there is one, otherwise to the nearest citation following the quotation within 260 characters, otherwise to the nearest preceding one. Each is checked verbatim against the cited opinion’s text from a local-first chain of public sources; a quotation is *near miss* when every fragment reaches token coverage of at least 0.85 without matching exactly, and *inaccurate* otherwise. The strict rate for a draft d is

$$S_d = \frac{n_{\text{accurate}}}{n_{\text{accurate}} + n_{\text{near miss}} + n_{\text{inaccurate}}}, \quad (1)$$

and a lenient rate counting near misses as correct is reported alongside it. Before any measurement the quotation checker was tested on twenty synthetic drafts with planted genuine quotations, wrong-attribution quotations, single-word corruptions, fabrications, and formatting artifacts, scoring 100 of 100; because that is a self-authored test, Section 6 also reports a provenance audit of the checker’s verdicts on real drafts. Pinpoint pages are verified against the official reporter pagination parsed from the Caselaw Access Project’s HTML volumes, which mark every page break; a quotation counts as on the cited page when its longest fragment lies within that page’s text, and adjacent-page matches are reported separately. Vendor citations (Westlaw and Lexis) are excluded from fabrication counts.

Human reference. The same instrument scored 482 clean, pre-ChatGPT federal appellate brief excerpts from the benchmark of Liu et al. [3]. Human lawyers reach 0.407 strict and 0.680 lenient quotation accuracy (578 scored quotations; 27 percent of scored quotations are near misses, chiefly bracketed alterations and ellipses) and 0.931 citation existence, the shortfall being index coverage of state and unpublished reporters rather than fabrication. The comparison is imperfect: human briefs cite more non-federal au-

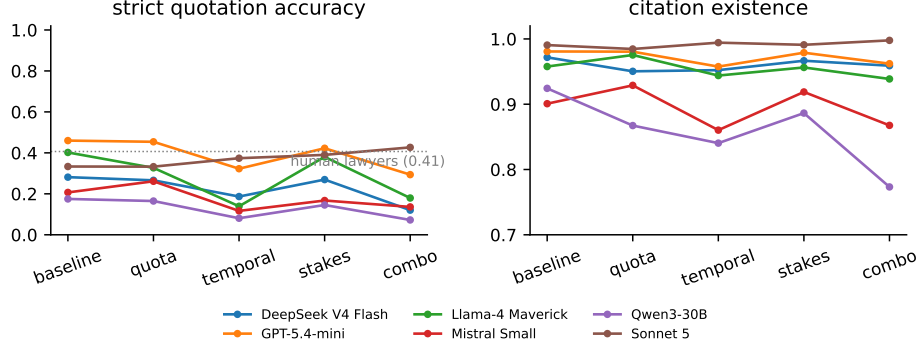


Figure 1. Strict quotation accuracy (left) and citation existence (right) by condition, one line per model. The dotted line is the human-lawyer reference under the identical instrument. The right panel’s axis starts at 0.7.

Table 1. Baseline versus the combined condition, rows ordered by baseline strict quotation accuracy. Quotation columns give the strict rate with the lenient rate in parentheses; existence columns give the rate with the raw count of citations not found in brackets. An asterisk marks a contrast significant after Holm correction within model.

Model	Quotes, base	Quotes, comb.	Exist, base	Exist, comb.	Temporal viol.
Qwen3-30B	0.175 (0.564)	0.072 (0.320)*	0.924 [28]	0.773 [90]*	20%
Mistral Small	0.207 (0.608)	0.136 (0.383)	0.901 [37]	0.868 [69]	35%
DeepSeek V4 Flash	0.281 (0.631)	0.120 (0.436)*	0.972 [10]	0.959 [20]	11%
Sonnet 5	0.333 (0.622)	0.427 (0.642)	0.991 [3]	0.998 [1]	0%
Llama-4 Maverick	0.402 (0.618)	0.179 (0.453)*	0.958 [9]	0.939 [31]	10%
GPT-5.4-mini	0.460 (0.722)	0.294 (0.571)	0.981 [8]	0.962 [27]	13%

thority and quote statutes and records more often than the models do. The human figures serve only as a reference for the severity of the verbatim standard.

Inference. The primary analysis is a citation-level logistic generalized estimating equation per model, $\text{logit}P(y_{imk} = 1) = \beta_0 + \beta_k$ for citation i in matter m under condition k , with exchangeable working correlation within matters [29], so that the 48 matters rather than the citations are the independent units. Each model has 8 contrasts (four conditions, two outcomes); we report Holm-corrected p values within model and call an effect significant only when it survives correction. Existence cells with few failures are reported as raw counts. A matter-level cluster bootstrap over draft rates agreed in sign with every corrected result. Hypotheses and decision rules were frozen before the full run; every later analysis decision, including two findings withdrawn after our own verification passes, is recorded in a forking-paths ledger released with the code.

4. Experiment 1: Pressure

Table 1 and Figure 1 report the rates.

Citation existence. Only the 30B model’s existence moves under pressure: from 0.924 at baseline to 0.773 under the combined condition, with the quota, temporal, and combined contrasts all significant after correction (combined OR 0.30, corrected $p <$

0.001). Mistral Small’s existence sits between 0.86 and 0.93 across conditions and moves in the same direction without reaching significance; the four other models stay between 0.94 and 1.00 in every condition.

Quotation accuracy. The temporal restriction is the condition that damages quoting. After correction, strict quotation accuracy falls significantly under temporal and combined pressure for Qwen3-30B (0.175 to 0.072) and Llama-4 (0.402 to 0.179), and under the combined condition for DeepSeek (0.281 to 0.120); GPT-5.4-mini and Mistral Small fall in the same direction without surviving correction, and Sonnet 5 does not fall at all. The Llama-4 row is the most consequential for practice: at baseline it quotes at the human reference level, 0.402 against 0.407, and under the temporal restriction its accuracy drops to 0.139 (OR 0.20, corrected $p < 0.001$). The lenient rates in Table 1 preserve the ordering, so the result does not depend on how bracketed alterations are scored.

An exploratory contrast at the top. Sonnet 5’s strict quotation accuracy is numerically highest under the combined condition (0.427 against 0.333 at baseline; uncorrected $p = 0.0292$, corrected $p = 0.233$), and its citations not found went from 3 of 315 at baseline to 1 of 449. Neither survives correction, the stakes condition alone moves neither outcome, and the temporal restriction changes the population of cited cases (median decision year 1943 under the combined condition against 1986 at baseline), so a shift toward well-memorized older opinions is an untested alternative to any interpretation in terms of more conservative citing. We report the contrast because it appeared in the pre-registered analysis and draw no conclusion from it.

Compliance with the temporal rule. The share of checked citations that violate the pre-1970 restriction is 20 percent for Qwen3-30B, 35 percent for Mistral Small, between 11 and 13 percent for the middle models, and 0 percent for Sonnet 5 (one violation in 802 checked citations). Compliance improves with capability except for Mistral Small, which violates the rule most while its existence and quotation figures sit mid-table; so rule compliance does not track existence or quotation accuracy across models. Zhao et al. [5] found near-zero violations with fabrication filling the window; here the smaller models both fabricate and violate.

Refusals and one model that stalls. Across the 1,152 pressured drafts there were three refusals (two from DeepSeek, one from GPT-5.4-mini). GLM-4.7-flash produced no output on 30 of its 240 drafts, 23 of them in the combined condition, reasoning until a 30,000-token budget was exhausted in two escalating attempts. Because its missing drafts concentrate in the condition under comparison it is excluded from Table 1 and reported here as a failure mode the taxonomy should include. Under the combined constraints this model produced no output in a substantial share of drafts, which is distinct from fabrication and from refusal.

Robustness to phrasing. The paraphrase arm was run on the four models available when it was designed. Under alternative phrasings of every condition on twelve matters, combined-condition existence was 0.885 for Qwen3-30B, 0.963 for DeepSeek, 1.000 for GPT-5.4-mini, and 1.000 for Sonnet 5, against 0.830, 0.984, 0.963, and 1.000 on the same matters under the original phrasing. No existence result changes sign; quotation rates on twelve matters are too few to test and are given in the released data.

Table 2. Revision under true checker feedback: scored quotations and accurate quotations before and after revision, strict accuracy, and episodes (of 24) that ended on a clean draft.

Model	Quotes scored		Accurate		Strict	Clean
	r0	final	r0	final	r0 to final	
Qwen3-30B	182	181	16	16	0.103 to 0.112	0
Mistral Small	145	133	15	17	0.208 to 0.264	9
DeepSeek V4 Flash	211	47	28	19	0.181 to 0.645	21
Sonnet 5	157	73	68	70	0.471 to 0.974	23
Llama-4 Maverick	64	29	11	8	0.176 to 0.349	21
GPT-5.4-mini	65	39	28	29	0.360 to 0.719	22

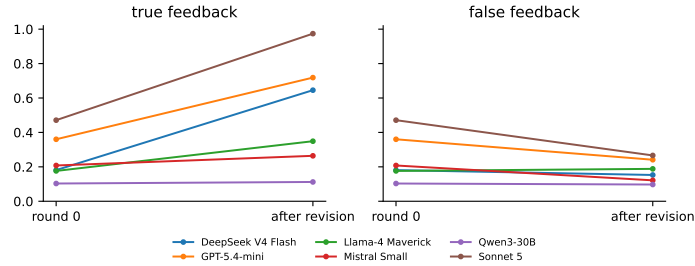


Figure 2. Strict quotation accuracy from round 0 to the final draft under true feedback (solid) and false feedback (dashed), one color per model. A rate can rise because accurate quotations were added or because inaccurate ones were removed; Table 2 separates the two.

5. Experiment 2: The Checker Inside the Revision Loop

Combined-condition drafts entered a revision loop. Each round, the checker returned one line per failure (a citation that does not resolve; a quotation that does not appear verbatim in the cited opinion), the model revised, and the loop ended on a clean draft or after three revisions. A control with *false feedback* gave the same number of identically formatted lines pointed at items that were correct. The loop used the first 24 matters in the pre-registered order (twelve per side). Loop rates are means over drafts, so round-0 figures differ slightly from the pooled rates of Table 1.

Satisfying the checker by deletion. Under true feedback the strict rate rises with capability, from 0.112 for Qwen3-30B to 0.974 for Sonnet 5, and the rate alone is consistent with repair. The counts in Table 2 show that the rise comes from deletion. Sonnet 5 went from 157 scored quotations to 73 while its accurate quotations went from 68 to 70; the rate rose because the number of scored quotations fell. Told that a quotation was not verbatim, the model removed the quotation, or its quotation marks, far more often than it corrected the words. GPT-5.4-mini and DeepSeek show the same pattern. Qwen3-30B changed nothing: 182 scored quotations became 181, and 16 accurate ones stayed at 16. Citation existence, by contrast, was repaired rather than avoided: citation counts were flat (220 to 220 for Sonnet 5, 198 to 197 for Qwen3-30B) while existence rose, so unresolvable citations were replaced with real ones. With respect to the second question in Section 1: at every capability tested, checker feedback led models to remove flagged quotations or convert them to unmarked paraphrase far more often than to correct them.

Neither outcome is flagged by the checker, and the revised draft does not indicate which claims were originally quoted.

False feedback. Under false feedback, four of six models ended with lower strict accuracy than they started, and the two weakest were unchanged within noise (Qwen3-30B 0.103 to 0.097, Llama-4 0.176 to 0.188). The strongest models were harmed most: Sonnet 5 fell from 0.471 to 0.266 and its accurate quotations from 68 to 26; GPT-5.4-mini fell from 0.360 to 0.241. When a correct quotation was flagged as inaccurate, the more capable models revised it, and the revision was usually inaccurate. Our checker’s own false-alarm rate on Sonnet 5’s drafts was 8 verdicts in 240 drafts, low enough not to trigger this effect; a detector with a false-positive rate in the tens of percent would be expected to produce the effect; intermediate rates were not tested.

6. Residual Errors in the Strongest Models

Three error classes remain when fabrication is rare, each measured deterministically on all six models.

Paraphrase inside quotation marks. Decomposing every inaccurate quotation against the cited opinion’s own text, 55 percent of Sonnet 5’s inaccurate quotations (413 of them) are recognizable paraphrases of the correct case presented as verbatim quotation, 202 match nothing substantial in the cited opinion, and 131 cite authority outside the corpus’s scope; GPT-5.4-mini’s split is 52 percent paraphrase. For the strongest models the most common inaccurate quotation is a paraphrase of the cited opinion presented inside quotation marks.

Real language, wrong case. Searching every inaccurate quotation against a segmented corpus of 8,392 Supreme Court opinions locates 463 quotations across the six models (114 for Sonnet 5, 66 for Qwen3-30B) whose distinctive language appears verbatim in a different opinion than the one cited. Because the attribution rule could itself misassign a quotation, the same search serves as an audit of the checker: it classified 8 of Sonnet 5’s inaccurate verdicts and 6 of Qwen3-30B’s as the checker’s own error, where the language was in the attributed case after all. A further screen for models reproducing the argued case’s own later-decided opinion from memory found 41 candidates, of which 32 were record or statute text present in the prompt and the rest doctrinal formulas that the argued opinions were themselves quoting; the finding was withdrawn and is recorded in the ledger.

Right case, wrong page. Where a pinpoint and a quotation co-occur on a U.S. Reports authority, the quotation lies on the cited page 77 percent of the time for Sonnet 5, 51 percent for GPT-5.4-mini, 58 percent for DeepSeek, 64 percent for Llama-4, 42 percent for Mistral Small, and 58 percent for Qwen3-30B, with pins outside the cited case rare at every tier. This measures, from public data, the error class on which published detection agents have the lowest recall [3].

Relevance judging: three failed calibrations. Whether a real, correctly quoted case supports the proposition it is cited for cannot be looked up. We tried to calibrate language-model panels against the expert-labeled content-misrepresentation items of Liu et al. [3] with a pre-declared bar of 0.75 accuracy on decisive verdicts: a three-model budget panel scored 0.500 with a binary verdict, 0.700 with a three-way verdict and abstention (misrepresentation recall 2 of 11), and a frontier panel of GPT-5.4, Sonnet 5,

and Gemini 2.5 Pro with 15,000-character grounded excerpts scored 0.594 (recall 4 of 17). No measurement verdict from a failed panel was interpreted; relevance is a stated limitation of this work.

7. Discussion

Whether a model fabricates legal citations under realistic prompting depends on the model. The lower-cost models are the ones most affected: for the 30B model a quota and an authority restriction triple the odds of a nonexistent citation. Quotation accuracy is more fragile than existence at every tier but the top, and the temporal restriction is the condition that degrades it most; restrictions on authority by date or jurisdiction are plausible drafting instructions, so this fragility is likely to be met in use.

Two practical conclusions follow from the revision experiment. Feeding deterministic checker output back into revision reduced the number of quotations and corrected few of them. A brief with fewer verbatim claims exposes fewer false ones, which has some value to a supervising attorney, but the remaining quotations have not been verified as corrected, and a reader of the revised brief cannot tell which claims were demoted to paraphrase. The checker is therefore more useful as a gate on output, holding a draft until its citations resolve and its quotations match, than as feedback, since revision in response to feedback removed quotations more often than it corrected them. Any system that feeds a probabilistic detector's flags to a capable model should measure the detector's false-positive rate first, because in our data the models whose accuracy rose most under true feedback also lost the most under false feedback.

The released verifier is usable by courts and regulators independently of the findings. Existence, verbatim quotation, and pinpoint page can be checked deterministically from public data, and the released verifier does so; court orders that require disclosure of AI assistance, of which the tracker records many [2], could additionally require the checker's output on the filing. What cannot yet be checked is whether a real citation supports the proposition, and none of the language-model panels we calibrated met the pre-declared accuracy bar of 0.75 on that question. With retrieval-grounded tools we expect existence failures to shrink further and the residual classes, paraphrase in quotation marks and wrong pages, to remain, since they arise in composition rather than recall.

8. Limitations

The study covers a single domain and task: Supreme Court merits argument, in one packet format, with one fixed template per condition and a phrasing check on twelve matters in place of a full paraphrase factorial. The task is closed-book, which matches the sanctioned filings but not commercial legal tools. The stakes clause names a sanctions rule that does not govern the forum. The strict verbatim standard penalizes bracketed alterations and ellipses for humans and models alike, and the human reference differs from the model drafts in citation mix. The provenance corpus covers 8,392 Supreme Court opinions and cannot adjudicate quotations from other courts. The false-feedback control is all false; intermediate false-positive rates were not tested. A no-feedback revision control was run after the pre-registered analysis and is reported in the released data rather

than here. Inference is per model at 48 matters, the independent units the clustering respects, so effects of a few points are not detectable, and the six-model set is one model per vendor rather than a sampled population, so capability, family, and training recipe are confounded.

9. Conclusion

Under prompting constraints of the kind found in practice, citation integrity degrades from the smallest model upward: nonexistent citations at 30B, misquotation through the middle of the range, and misattribution and wrong pages at the top. A deterministic checker built from public data detects each of these error classes. When its output is fed back into revision, the revised briefs contain fewer quotations and few corrected ones, and false feedback lowers the accuracy of the models that revise in response to it. The instruments, the seeded-fault validation harness, the forking-paths ledger, and per-citation records for every table are released so that the checks can be run on any brief.

10. Ethics and Release

All source material is public court data. Drafts were generated for measurement and none was filed. Code, instruments, drafts, and records are at github.com/hardy30894/citation-under-pressure. Language-model assistance was used in writing code and prose under the author's direction; every reported number is produced by the released deterministic instruments.

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